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H8SCOOP

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BITS & PIECES

H8SCOOP MAIL SCHEDULE

The next few H8SCOOP's will be mailed early. This is to avoid the December Christmas rush of mail, which will not see any H8SCOOP's this year. Last year there were too many problems with this newsletter either not getting there or being delayed. This NOVEMBER issue will be mailed mid October, the DECEMBER issue will be mailed mid November, and the January issue will be mailed early January. This way there will be no December mail and all mail problems should be avoided.

400K UPDATES--NEW SOFTWARE

I have received the latest version of the SY.DVD driver from Dean Gibson of UltiMeth Corporation (see **NEW STUFF**). This version is an improved version and will sell for \$40. Those having the old version can send me the original disc and \$20 and I will send you the update. This price will include the supplying of all written documentation. This price was set per request of Dean Gibson, the author.

The Double density driver for the H89/Magnolia board is also available from myself for \$40. This will allow HDOS at 650K for those having the Magnolia double density board in their H89 (I sell this with cables and CP/M for \$510) and CP/M at 760K.

I now have the modifications for the Heath CP/M BIOS for \$25. This modification to be installed includes full instructions and will allow you to run CP/M for about 390K on the single density unmodified H8 or H89 with the Tandon drives. Note H8 must still have side select function derived somehow, and beware of the circuit in REMark 18 as I have had many complaints about it not functioning properly.

I have also lowered the price of my drives and software packages. The reason for doing so were twofold. First, I again got a price break because of the volume of drives I have handled in the past. Also I noticed others offering the drives for lower prices. Most give absolutely NO support for this, so if there are problems, who do you have to help you get up and keep you going? No one. I want to make a point perfectly clear. I was on the bottom of this entire 400K operation, and now individuals are trying to move into this territory. That's fine, that's what free enterprise is all about. But NO ONE can offer the support I can, and I'd like to keep it that way. Anyone who has purchased the drives or packages from me has received full support and will continue to do so. This applies to new purchasers also. Don't try to save a buck and regret it. I offer unconditional 90 day warranty with new drive replacement if any problems arise, and after that, can refer you to a excellent quality low cost reputable drive repair service whom I have been more than satisfied with. As usual you will be kept up to date on all updates and developments concerning these drives and software. Because I was on the ground floor, you can be assured you will see it FIRST in H8SCOOP.

My new prices reflect the \$40 software package, and even at that I am lower than before! Single drives with the \$40 new software package will sell for \$465. Single drives alone for \$435. The second drive will sell for \$430, to quantities of 8 or more for \$410. And this is the price for FULL SUPPORT! I also carry commercial quality single cabinets for \$100 with power supply. Single 34 conductor cables sell for \$25. Complete maintenance manuals sell for \$25 plus \$3 for shipping and handling. All the other

prices include shipping--No hidden cost. Charge card orders add 3%. For more information, or further breakdown on prices, write for drive price information.

A note on these drives, when not in use, keep the door open! When the door is closed, the heads come together, and without a disc inserted, it's not too good. How would you like to bang your head against another? This is more important when shipping or transporting the drive. Wire the door open when shipping or transporting so the door cannot close.

. . AND OTHER DRIVES

Because many individuals are calling about other drives, I have decided to also distribute these. The Tandon TM-100-1 is a direct replacement for the standard Heath 100K drive. The best part is it will step at 5ms and doesn't have the head load solenoid, but loads the head with the door closed, just like the 400K drives. This means less noise. For those looking for replacement drives or wanting to configure their own systems, this is the one to get. I will sell this for \$230.

Another alternative which seems to be in demand as I have already sold several is the TM 100-2 which is the single density 40 track, but double side. This will do nicely as SY0: to maintain software compatibility, but yet give you the double sided option for 200K. It works nicely in the H89 and sells for \$340.

WHAT'S IN A NUMBER?

The suggested retail price list guide for Zenith Data Systems introduces some models and prices you may not be familiar with. The Z-89-1 for \$2,895 is the Z89 computer with Z19 terminal with 48K and a built in disk drive, 2 serial I/O ports. The Z-89-0 is the same but without built in disk for \$2,595. Here's a new one. The Z90-0 is the same as Z-89-0 except it utilizes the Z-89-37 double density controller and has 64K standard for \$2,895, available fall 1981. The Z-90-2 is the same as Z-89-1 but with double density controller for \$3,195. The ZVM-121 is the \$150 monochrome green monitor described in last month's issue. The Z89-6 is the upgrade kit to convert the Z89's to Z89-1 (formerly Z89-FA). It is required to attach an outboard disk drive unit to the Z89 or WH-89 and sells for \$50.

HMS-817-2(b) is MICROSOFT FORTRAN supplied on 5.25" discs for CP/M with 48K capacity for \$195. HMS-817--3(b) is MICROSOFT COBOL on 5.25" CP/M for \$395. HCM-817-1 is CBASIC for CP/M on 5.25" for \$115. HMP-817-4 is CP/M DATASTAR on 5.25" for \$295. The rest you should be familiar with from the Heath catalog, or appears on 8" discs which I won't mention since not many have them. The entire line of PEACHTREE business software is now available on 8" for \$395 each, which includes General Ledger, Accounts Receivable, Accounts Payable, Payroll, Inventory (\$495), Sales Invoicing (\$295), Property Management (\$995).

HEATH CHRISTMAS CATALOG OBSERVATIONS

Looking at the new Christmas catalog I am a bit puzzled. The double density controller board for the H89 is now available (as mentioned many issues ago), but with the Tandon drive (that's sure what it looks like at \$550 compared to my \$435) it only gets 640K. If that's all it gets, I'd be leary, as with the standard card and the new driver, we are getting 400K--not much of an increase. The hard disk is also available for \$5800 plus \$195 for interface. The odd thing is, it mentions no new HDOS to

support it, and I doubt if 2.0 will!

NEW BB

New is a Hayes smart modem for \$279, the Type-N-Talk for \$375, and yes, the H25 high speed dot matrix printer for \$1095 kit form, \$1595 assembled. Surprisingly they have added the MX-80 for \$545 plus \$65 for serial interface. The H14 was indeed reduced to \$395 kit form which I feel is a very good deal (and my local Heath manager told me that would never happen!).

In software, something we've been waiting for, the Microsoft BASIC compiler under HDOS for \$250. I hope it doesn't blow up program space like the FORTRAN did. A good deal of CP/M business software from Peachtree including General Ledger for \$395, Accounts Payable for \$395, Sales Invoicing for \$295, Accounts Receivable for \$395, inventory management for \$395, and rental properties for \$995. No mention yet of new CP/M release. Also under CP/M, DataStar, SuperSort, WordMaster, Condor Data Base Management, and yes, SuperCalc.

My guess is there may be great software compatibility problems with all these new controller formats, I'll review this next month as time and space permits.

PERSONAL COMPUTER MARKET

IBM's announcement into the personal computer market has everyone wondering. They may capture a good share of the market if they go about it right. However, IBM's past history shows service dropping from what it once was, and delivery usually far behind schedule. If this trend is to keep up for the personal computer, they'll not do as good as they want.

The estimated Personal computer market for 1981 is something like this. Apple has 23% sales, Tandy with 16%, Commodore with 10%, Hewlette-Packard with 6%, Nippon Electric with 7%, and all others with 38%, which Heath/Zenith is included. I wish I had a breakdown for the others, but it is not available. I have a feeling, the H/Z89 market is quite strong, although possibly not under that name. I have noticed lot of other companies selling word processors and computer systems with any other name except Heath/Zenith H89/Z89, although this is the backbone of the system. I feel if all these were counted, the Heath/Zenith line would be out of the classification "others" and into their own category.

HULLAND'S HDOS 0.1

A news release from Hulland Engineering to all users of HDOS 0.1. "A problem has been recently reported to me regarding the use of Heath's BASIC (version 2.0) with HDOS 0.1. The problem is that programs can not be LOADED or SAVED, and there is some danger of crashing the system if an attempt is made to do one of these operations. The problem occurs only with the 2.0 version (not 1.6 or Microsoft BASIC). It is caused by the fact that Heath's BASIC assumes that it "knows" the address of one of HDOS's variables. Unfortunately, that variable is at a different address in HDOS 2.0 than it is in either HDOS 1.6 or HDOS 0.1.

"There are two ways around this problem: 1) Use either Heath's BASIC version 1.6 or Microsoft BASIC. 2) Patch the Heath 2.0 BASIC for use with HDOS 0.1 as follows. Use PATCH, BASIC as the file name, and change address 112275 from 121 to 120. Note that PATCH can modify BASIC only after both programs have been CHNGVER'ed and that BASIC.ABS must not be write protected.

PIE-DESPOOL CP/M PROBLEM

There is a conflict between use of the PIE editor under Heath CP/M and the DESPOOL utility. Invoking PIE causes DESPOOL to unload, terminating any print operation in progress. PIE can be patched to avoid this problem. Anyone having that problem contact The Software Toolworks for the fixes.

The New Orleans Heath store has a corvus 10MB hard disk connected to their computer and it is available on a self written BB. The number is (504) 467-6323. To start, enter CTRL C. This is a CP/M system and uses standard CP/M operating procedures.

5MB DISK FOR H88/H89 UNDER \$2,500

Floppy Disk Services/ CN 5212/ Princeton, NJ 08540/ (609) 771-0374 has available a Technology 5MB hard disk controller setup complete, for \$2499. This includes a real time clock, and is all assembled and tested. Contact them for more info.

NEW STUFF

MORE FROM ULTIMETH

Ultimeth Corporation/ 24025 Fernlake Dr./ Harbor City, CA 90710 (213) 539-4276 now offers 4MHz software for the H8 and H89. All products will run at either 2 or 4MHz. If necessary, they measure the CPU speed and adjust their internal operation accordingly. Use of MTHREX/MCTOCT, DKH17V2.DVD, DKDDFLP.DVD at 4 MHz assumes that you have an H8 or H89 already functioning properly at 4MHz. They cannot provide hardware assistance in getting your computer to run at 4MHz, and will not answer mail requesting such assistance. Some of the recently published notes on H89 4MHz mods in newsletters and on MNET violate numerous CPU board chip specifications and will not be supported or condoned. In the future, UltiMeth may offer a reliable 4MHz mod for the H89.

MTRHEX and MTROCT ROMs which are replacement monitor ROM's for the Heathkit H89 computer provide increased HDOS processing speed and additional debugging capabilities. Additional features have been mentioned in past H8SCOOP's. This now also supports CPU speeds of 2 or 4 MHz. This sells for \$50, contact Ultimeth for more info on this one.

DKH17V2 is a replacement mini-floppy disk device driver for the H8 and H89's. It is a new version which exhibits all the old features as mentioned in past issues of this newsletter, but also provide the following new ones.

The head is unloaded during the disk-ready check used by the HDOS RESET command, allowing discs to be removed and inserted with the head unloaded (this is not such a concern for the Tandon drives, as they do unload the head whenever the door is opened).

The device driver also measures the speed of the CPU and computes the correct value for internal timing constants for a CPU running at either 2 or 4 MHz automatically.

The number of mini-floppy disk drives attached to the computer may be specified. Since HDOS 2.0 allocates 256 bytes of RAM for each possible drive, users with one or two drives will save RAM space.

Several Heath bugs are corrected or circumvented.

This sells for \$40 which includes written documentation and shipping to US addresses. Previous purchasers of DKH17 may exchange an old Distribution diskette for \$20 through myself of UltiMeth. Source code is NOT provided.

DKKKFLP.DVD is an HDOS 2.0 double-density floppy disk device driver for use with the H89 and Magnolia Microsystems double density board. It sells from myself or Dean of UltiMeth for \$40. This driver will support both the Magnolia and Heath's Single density boards at the same time.

Dean is now working with Trionyx Electronics in developing new software. In the future we should see some GOOD stuff coming from Trionyx if Dean's new device driver is any example.

MICRO MOONLIGHTER NEWSLETTER

J. Norman Goode/ 2115 Bernard Ave/ Nashville, TN 37212, offers MICRO MOONLIGHTER NEWSLETTER. It caters to those who wish to turn their investment in a personal computer into a business venture which will recover their system cost or even provide full time earnings for an at-home career. The newsletter contains no games or tutorial articles on computer basics as are usually found in the popular trade press. It does provide the latest techniques, tips, and guidance for the person who is 'comfortable' with his system and wishes to transform it into a money machine. Marketing methods, advertising hints, the ins and outs of contracts, and industry trends will be covered in regular departments.

Each issue features reviews of books helpful to the micro-entrepreneur in such areas as business start-up, mail order methods, tax shelters for the small businessman, and personal motivation. Emphasis will always be directed toward getting the greatest return from every hour and dollar invested.

Subscription rates are \$25 per 12 issue year in the US, \$29 Canada, and \$35 worldwide. VISA and MC welcome. Contact J. Norman Goode at above address for more info, or to subscribe. I want to add I reviewed some issues and found some good tips. Since it just recently started, I feel we can expect some good feedback and ideas from this newsletter for those wanting to make a few bucks. The only negative comment I have to make is there is lot of "white" space which could be used to print more useful information. The issues I have seen are 8 pages in length, same as H8SCOOP.

UPDATE ON REMREF

In last months issue I mentioned REMREF in this column. The author, David Powers, writes in some corrections to the article. He says although KEY-IT is available from Keyboard Studio and most Heath Stores, the actual REMREF is only available from himself (David Powers/ POB 1154/ Troy, MI 48099). He also mentions that although REMark may be printing a general index, this investment will allow one to avoid the painstaking typing in of all the entries, and for the price is well worth it. I have to agree with him on that, since it is 91 sectors in length, which translates to about 24,000 characters! The entire file can be processed in about 1 minute under CP/M compiled MBASIC, or about 3.5 minutes under HDOS MBASIC.

SOFTWARE FROM D-G ELECTRONICS

D-G Electronics/ 1827 S. Armstrong/ Denison, TX 75020/ (214) 465-7805 has added some good software products to their ever popular hardware line. The have been known for their H8 hardware products which include Z80 4MHz CPU, FP8 monitor which supports CP/M and HDOS, 64K Dynamic RAM, CMD-1 which is their answer to the 0-org (not required for use with their 64K memory board) and ADP4 which allows for 4MHz operation. For a pamphlet on these items, contact DG. They sent me a note that the FP8 monitor package is now included in the \$249 price of the DG-80 CPU, which represents a \$69 savings.

In software packages, **ARCHIVE** is a disk archive utility for use with more than 2 drives allowing you to store, retrieve and directory the contents of a complete diskette into a single save file. Utilizing a fact that files are set up by groups, this utility allows the user to waste at most one group on the destination device. This is useful for larger discs such as the 400K Tandon where the group size is 8. To copy small 1 or 2 sector files individually would

normally take up minimum of 8 sectors each. With **ARCHIVE**, many files can be combined into one for saving purposes, thus saving this wasted space. Similar in concept to idea mentioned in H8SCOOP, page 1, September. For use with HDOS 2.0. Price \$49.95 on 5 1/4" disc or 8" disc, add \$40 for source code. All software is supplied on your choice 5 1/4" or 8" discs.

PRELOAD is a Pre-Load absolute binary module for \$29.95 (add \$20 source) which allows the user to load in absolute binary files (.ABS) into an auxillary RAM area. This is usually a second or third DG-64D 64K RAM card in the H8. In this way, memory is used as a peripheral mass storage device. When a program is linked to, via the .LINK SCALL, PLM will intercept the call and check the internal directory for the named file. If found, it will be loaded and executed, if not found, it will be passed on to HDOS.

DISK MANAGEMENT UTILITY PACKAGE for \$49.94 (add \$40 for source code) is a package which supports all drives, including the 400K Tandons! It is comprised of three programs, PRODUMP, DSKCPY, and BAD.

PRODUMP is an intelligent Disk Dump Utility for HDOS 2.0 allowing you to dump from any device created for the H8/H89. Like a dump program, you can edit files on sector by sector basis, but don't have to remember sector numbers. All output can be directed to Hardcopy device. You can teach PRODUMP about new directory devices, so it can grow with you. It will support standard 40 track drives, 8" drives, and the 400K Tandons, plus others. An excellent feature is the ability to search a file or disk for any string in ASCII or the current mode which is set. An HDOS disk can also be accessed by group numbers without sector calculations. Modes supported are OCTAL, HEX, or DECIMAL, plus the display in ASCII.

DSKCPY is a Universal Disk Copy Program for HDOS 2.0, which eliminates the need for device dependent copy programs such as FTCOPY, COPYER, DUP, etc. It also runs faster than most of the others. All devices are supported, including the 400K Tandons, the H47 and of course, the standard 40 track drives. Non-HDOS volumes such as CP/M can also be copied with this utility. Volume ID's and Labels can be changed on the destination.

BAD is a Bad Block Correction Utility. It can be used with any HDOS recognized directory device (yes, includes the Tandon 400K) to remove bad blocks. By telling HDOS that the group is bad, the user will save the frustration of having HDOS pick a bad sector and return a "Write Failure" error message. By using BAD the user may keep any disk up to par. By reading each group on the disk, the user may remove sectors from a file which has bad sectors, then save most of the data contained in the file without getting a "Read Failure" error message while copying the file. A user can also tell before an INIT if there are bad sectors which should be reported to INIT to log them as bad. Again, hard copy reporting is available on bad blocks found. Other features are ability to test multiple disks without exiting from the program, ability to remove groups from any part of the file, or from the free pool.

SYSCMD/plus for \$39.95 (source code \$30) is an extended system command processor designed and written to add a new level of power and flexibility to the HDOS operating system. All standard commands you may be used to still apply, which makes transition less frustrating. Many commands are abbreviated, and new ones added. Special features such as MULTI-TASKING operation is supported. It is accomplished through the use of special interrupt driven programs, known as TASKS, in conjunction with the Task Manager, which is provided. You can also write your own tasks to perform home security, clock (one provided), print anything appearing on screen (provided), and more. The installation of SYSCMD/plus is easy, and allows you to customize it to your wants.

CP/M MICRONET PROGRAM

Tom Jorgenson of EasySoft (CP-MIG) writes in "Micronet has licensed all three CP-MIG sysops (CP/M Interest Group on Micronet) to distribute copies of the CP/M version of the Micronet Executive (MNEEXEC) program. This is a modem utility which is intended for file transfers with Micronet with handshaking and error-detection. The utility was written by Russ Ranshaw of the Compuserve staff.

"Once this utility has been adapted for the H8 and H89 it will be made available in three ways. First, those who presently have the ability to transfer files from the net will be able to download the required files directly from the CP-MIG database area. Secondly, the package is being submitted to HUG without charge. Thirdly, EasySoft will distribute the source code for a handling charge of \$10 for Heathkit code, or \$15 for the full CP/M set (full set is 2 disks). These will be on standard H89 5.25" format.

If you are interested in this, contact Tom of EasySoft/ 122 Yankee Drive/ St. Charles, MO 63301.

HAM STUFF FROM COMMSOFT

Howard Nurse of COMMSOFT/ 665 Maybell Av/ Palo Alto, CA 94306 (415) 493-2184 sends along two press releases describing the latest software from Commsoft for the H8/H19 or H89.

"Hams who own Heath computers can send and receive Morse code with the new CW89 software package from Commsoft. This featured packed program includes a split screen display, 4-99 WPM operation, receive autotrack, a 1000 character pretype buffer, 10 user-definable messages, unique break-in mode, on-screen system status, disk I/O, hard copy and a versatile code practice section.

"CW89 allows the user to practice code at variable speeds in several formats--alphabet only, alphabet or numbers, or all common Morse characters may be sent with or without random spaces. Practice code can come from one of 100 disk files which can be prepared using a standard text editor. All communications and practice text can be sent to a printer or stored on disk."

The CW89 program runs under HDOS with one drive and 32K RAM. A hardware interface such as COMMSOFT CODEM is also required. CW89 is \$99.95 postpaid. A complete package consisting of CW89, the CODEM, a computer interconnect cable, power supply, complete documentation and shipping is available for \$249.95. California residents add applicable sales tax. VISA and MC accepted.

The COMMSOFT CODEM, a universal CW interface for personal computers is now available for radio amateurs. The CODEM provides an easy way to get your Morse code software on the air. Converting received CW audio to RS232 or TTL signals and RS232 or TTL signals to transmitter keying, the CODEM doubles as a code practice oscillator and CW regenerator.

A sharp 800 Hz bandpass filter, AM detector and low pass filter are designed into the CODEM to provide outstanding noise and QRM rejection. CW can be monitored using an internal 2" speaker or with an external high impedance earphone. Front panel sensitivity, tone and volume controls are provided. The CODEM comes with a comprehensive manual which includes operating details and connection instructions. It requires an external 9V power supply and costs \$124.95, with the power supply available for \$9.95. Add \$5 shipping and handling.

FOUR DRIVE HDOS!

Four drive HDOS! How is this possible? All you need is one IC and the modified device driver. In fact, up to 7 drives

can be selected by using this modification. Unfortunately until the H17 ROM is modified, only 4 drives can be used.

With this modification all versions of CP/M and HDOS will still run normally and it will not affect operation with the H47 disk system. The modified files for both the Heath SY: and the HUG SY are included along with instructions on how to assemble the new SY:. Instructions are also included on how to configure SY3:, for drives by Wangco, Siemens, or Tandon. Runs on H8/H89 HDOS 2.0. Send \$18.00 or \$15 and disk, to Gregory Marsh/ 5279 Miles Ct/ Woodbridge, VA 22193. Specify 40/80 track, 1 or 2 sides.

OFFERS

This month's offer is MULTI-KEY sort for \$6. For this price it will be supplied on an excellent, but used diskette! Since going to 400K drives, I have recovered hundreds of discs which were used only once, to put the information on them for storage. This sort was submitted by Fred Pospeschil, and is the best sort program I have seen thus far.

MULTI-KEY sort was written in C80 and comes complete with all source listings. All data and code is provided to recompile the program and make modifications, should you so desire. Because it is written in C80, an ABS code is generated which is machine executable and can be used in systems with as little as 16K and one drive.

Features in the program include the following: 25 sort fields which can be one character to the full length of the record, sorting on ascending and descending directions. Up to 32000 fixed or variable length records each of which can be up to 32000 characters can be sorted, but the upper limits cannot be realized because of computer system memory limits. Sort fields can be located anywhere within the records. Terminal independent.

An excellent program, at this price, you can buy several and give one to the wife for Christmas! The cumulative OFFERS will be printed in the January issue.

REQUESTS

ZEN KUROKAWA/ 1469 12th Av/ San Francisco, CA 94122 would like to know if anybody has used the DYNATYPER sold by Rochester Data Inc to drive a Selectric typewriter with their H8 computer. He wants to use his IBM Correcting Selectric as an output device, but wants to know if anybody has done this using the Dynatyper interface. If anyone has info on this, please contact Zen.

There have been several requests on using the H9's and other terminal's as remote devices. This would require an RS-232 EXTENDER device, as RS-232 signals are generally not good over about 10 feet or so. This device would have to transmit and receive at least 100 feet. I imagine an RS-232 driver device would do the trick to be able to send signals the extra distance to remotely locate terminals using RS-232. I realize the normal way of doing this is to use a 20mA loop, but that's not too available in the Heath systems. Anyone having ideas on this, please contact H8SCOOP.

I am working in getting up a LUCIDATA PASCAL LIBRARY, anyone having programs to donate or swap, please let me know. This library will be available to all subscribers once we have some programs in it.--H8SCOOP.

Jan Johnsen/ POB 993/ Bellevue, WA 98009 wants to know if anybody has used a modem to transmit UCSD Pascal files. I know the Lucidata uses standard ASCII files created by any text editor, but UCSD has its own editor. He wants to be able to send Pascal ASCII text from one

unit to another. While on the subject of this Pascal, he also wants to know if anyone has figured out how to adapt it to work with the 400K drives. Any info on this, contact Jan and H8SCOOP as I will make it available to all. It seems that others have this same request. Bill Gunckel also wants a patch or whatever is needed to allow the 400K drives to be useable with the UCSD Pascal. I fear since this package uses its own disk device driver, it will be a major undertaking.

GEORGES TROUPIN/ 22 Rue De Grady/ 4920 Embourg/ Belgium has the H8 extended configuration option to run CP/M 0-org. His problem is his former 56K system lost 8K when running HDOS. He has tried the HDOS 1.6 patches for 64K ram published in BUSS #32 but without any success for both of them. If anyone has any patch to recover the lost 8K, it would be a great help. The only one I know of for 1.6 is to out 000.362 before pushing GO to BOOT. You can not get the full 64K under 1.6 or 2.0, as this is still used at the low 8K end, the system will still sign on with 56K RAM. 1.6 will not work without this 000.362 OUT-GO routine though.

ROBERT GREGORY/ 12600 S. 73rd Av/ Palos Heights, IL 60463/ (312) 448-0842 is having problems interfacing his H8-4 interface card to an Integral Data Systems IP-225 printer. It appears that either the interface card or device driver software (using the LP:) is not recognizing the CTS handshake signal provided by the printer. It successfully runs at 600 baud, but not the recommended 1200 baud or higher. Anything over 600 causes over run of lines being printed. If anyone out there has interfaced this printer, or knows what the problem may be, contact Robert.

RAYMOND PADILLA/ 708 Emmet/ Ypsilanti, MI 48197/ (313) 483-5765 wants to know if anyone has modified any of the Heath device drivers (or written one) to handle a DEC LA-120 terminal at 9600 baud. He is having trouble with handshaking at this speed. He also wants to know if anyone has written a program for estimating race track results. Please share the wealth--even for a fee!

W H O ' S W H O

NOTE 3 FOR H8 GRAPHICS BOARD OWNERS

Fred Pospeschil/ 3108 Jackson St./ Bellevue, NE 68005/ (402) 291-0795 has another note for users of the HA-8-3 graphics board.

"Over the past few months I have had the opportunity to test and review the Lucidata Pascal (from Larry Reeve of Polybytes) graphics compiler I mentioned in an earlier note. This compiler greatly simplifies programming the color graphics board. Everything you need to do can be done from within the Pascal code without resorting to external user written ASM routines. Like UCSD, Lucidata is a P-code system and thus does not run as fast as Tiny Pascal but gives you much more of the language to work with. Using the Lucidata graphics compiler makes the program "JIM" (now available through HUG) run almost as fast as the Tiny Pascal version.

"Speed is not the whole story. The compiler provides over 45 built-in functions which can be called to control and set up all the hardware on the HA-8-3. In addition to provide easy control over the board functions, Larry has spent considerable time developing extended functions which are both difficult to program and slow to execute if done in P-CODE. These functions include:

- Read an analog Channel
- Initialize the graphics chip
- Move to a specified MC or G2 mode X/Y point
- Draw/erase a line to a specified MC or G2 mode X/Y point
- Define/set regional G2 colors
- Fill/clear an area on the screen
- Define, update and move sprites
- Move blocks of data between Pascal and VRAM

"The line drawing/erasing routines, written in ASM, use efficient algorithms, similar to the ones used by HP and Vectorgraphics. The sprite pattern definition function is beautiful, you can actually see the sprite pattern in your program source code.

"The draft user documentation for these graphics functions runs about thirty pages and contains a HACOMMON.PAS file, akin to the HDOS.ACM files, to keep all of the definitions and references straight. The compiler is compatible with both the 9918 and 9918A and a separate version will also be provided for those with the ADM9511 math chip installed on the board. The math chip version, which will be supplied at no additional cost, uses the chip version, for all floating point and mostly integer arithmetic functions and allows direct access to the extended functions provided by the chip.

"Using this compiler doesn't eliminate the need to read and study the basic manuals for the VDP and PSG chips but it sure makes things easier once you have. It looks like the system will be distributed on two disks accompanied by a substantial user's manual of over 100 pages. The price will be \$105 to first time purchasers, but only \$30 for standard system owners. I might add that the non-graphics compiler is also a cost effective item and we now have twenty copies purchased for our local club members. Those who have gotten into the package have favorable comments and have had no complaints as yet."

* PROGRAMMING GOODIES *

A good rounding routine for MBASIC is listed as follows. Variable A# must contain Dollar and cents amount to be rounded upon entry, and will contain rounded value upon exit. This will round to nearest penny for both positive and negative numbers. Thanks to John Welch for this one.

```
A#=100*A
A#=SGN(A#)*INT(ABS(A#)+.5)/100
```

Try this one out, also from John. To use the string functions in MBASIC try

```
DEF FN$(A$)=LEFT$(A$,2)+"**"
```

and latter print FN\$("HI THERE")

Neet. Very few BASIC's have this feature.

-This one by Pete Swindall from August 1981 CHUG. "Here are some tips on printing graphics and interfacing to the H19/H89 terminal from Microsoft (r) Basic. Knowing these undocumented facts may save you quite a bit of aggravation.

(1) Whenever you send a master-reset to the terminal (ctrl Z) always be sure to follow it immediately with a brief 'null' time delay; i.e., FOR I=1 TO 10: NEXT I. No terminal I have encountered can perform a master-reset fast enough that it doesn't lose several characters immediately following.

(2) Any time you are doing a graphic display that requires 80 or more characters on a line, it is good form to initialize the terminal by sending 'CHR\$(27);"w";'. This will cause the overflow to be discarded and not force an extra blank line for people who have line length set to 255 and WRAP turned on."

Russ Scholtz has made this CP/M patch for comet/starwriter printers. Use HDOS DUMP. Change byte F2 of sector 047 from C2 to CA (BIOS.SYS on 2.2 distribution disk). This reverses the logic necessary for the Comet to run on CP/M.

TECHNICAL FORUM

ON Z89 to H19 TERMINAL MOD

Bob Sloat writes in an idea for me which will allow the Z89 to be used either as a Z89 or H19 video terminal. This would be especially useful to those who have or want an H8 but don't want an extra terminal sitting around. Bob says P-404 of the H89 can be interchanged with P603. This makes the CPU logic board inoperative and the terminal logic board is connected to the back panel plug A, the DCE cable to the Serial Interface Board where P603 plugs in. It seems, Bob says, that a 2 pole multi terminal rotary switch could be wired to eliminate the need to open the H89 and having to remove several boards to make the change and replace the boards to use an H89 as an H19 terminal.

H19 AT 9600 BAUD

Chris Simmons writes in about all the commotion about the H19 running at 19200. For those applications where the switch settings will not give you 19200, the following may help.

Remove the Terminal Logic Board of the H19. Locate JP10 jumper and the lower left of the board, and cut the hardware jumper between 5 and 2. Install a jumper from 5 and 1. Reinstall the board, and the TPU now runs at 3.057 MHz. It is important to note, however, that all CPU's can't handle this increased speed, and you may have to either replace the Z80 with a Z80A or reprogram 5 to 2 to go back to original configuration. SW #401 is set (left to right) 0100. He wishes to thank Roger Sullivan for that info.

Next refer to H8SCOOP issues May and June 1980 for setting either the switches or setting the escape sequence to run the terminal. Chris says he has had no problems with his terminal after the mod. He wishes to thank Roger Kulstad, Ted Mieske, and D. C. Shoemaker for that info. Also note that after the mod, if you have discs set at 9600 they will no longer come up. He wants to know if anybody knows how to change the baud rate and where on the disk it is located.

PRINTERS AND PROTOCOLS--CP/M

The following information from Tor Tverre. "In the July issue of H8SCOOP Bruce Bennett asks for info on making his IDS 460 print faster than 300 baud under CP/M. Questions like this keep popping up. Without some knowledge of Assembler and printer protocols it is hard to understand what is really going on."

"When a computer wants a printer to print, it outputs characters to the printer through the printer port. If the characters arrive faster than the printer can print them, the print becomes garbled. To avoid this there has to be a way the printer can tell the computer that it cannot receive any more characters. This communication between printer and computer is termed 'printer protocol'. Unfortunately, there are many printer protocols, and if you are using the wrong one strange things will happen. The most common are XON/OFF and DTR (data terminal ready). Unfortunately Heath's printer drivers do not support any of these, and that is the reason for most of the trouble with interfacing printers to Heath computers using Heath's printer drivers.

"The Heath H-14 and H-24 printer drivers are both configured for the RTS (request to send) protocol. To rewrite a printer driver from RTS to the XON/OFF protocol is a major operation. The XON/OFF protocol requires the printer to send a CNTRL S to the printer to stop transmission and a CTRL Q to start (same thing you do with your H19 to start and stop scrolling).

"Before a printer driver is rewritten an understanding of the H8-4's 8250 ACE (same as the H89 serial interface) is necessary. The 8250 is a programmable Asynchronous

Communications Element. It has programmable baud rate and independently accessible control registers. The 8250 occupies 7 consecutive addresses, the main port and 6 control registers. The register containing RTS and DTR status info is #7 (address port +6). The registers contain eight bits, numbered from 0 to 7, the Least Significant Bit being bit 0. Bit 4 contains the RTS (or clear to send) information and bit 5 contains the DTR (or data set ready) information. The standard RS-232 interface assigns pin 4 to the RTS signal and pin 20 to the DTR.

"Let us take the H-14 as an example. When the printer buffer is nearly full the RTS (pin 4) is set high. This signals the computer to stop sending characters to the printer. When the printer is ready to receive more, it drops the RTS signal to a low. To make use of the signal stored in bit 4 of register 7, the printer driver must read this register, isolate bit 4 and check if it is low or high. This is done in Assembler as follows:

```
MOV     A,M      GET BASE ADDRESS (PORT+0)
ADI     6        ADD OFFSET TO POINT TO 8250
STATUS REG
CALL    PIN      LOAD REGISTER CONTENTS
ANI     010H     LOGICAL AND WITH XXX1XXXX
JNZ     LPTOS1   JUMP TO PRINTER BUSY ROUTINE
IF NOT ZERO
```

"If we want to rewrite this routine to a DTR protocol it will look like the following.

```
CALL    PIN      LOAD REGISTER CONTENTS
ANI     020H     LOGICAL AND WITH XX1XXXXX
JNZ     LPTOS1
```

The reason for ANDing the register with 10H or 20H is to isolate the bit we are interested in. If we are interested in looking at bit 5, and it with 20H

"Printers like the IDS 460 and 560 are dropping the DTR to prevent overflow of the buffer when full. This is the opposite of the H-14's RTS signal, which makes it necessary to change the JNZ instruction to JZ (jump if zero rather than jump if not zero). In other words, all that is necessary to patch the CP/M BIOS.SYS for the IDS 460 is to change two addresses. There is more than one way to do this, the simplest is to patch the BIOS using DDT and write it back to the disk. For those not quite familiar with CP/M, the following is recommended.

- 1) Mount a copy of your distribution disk.
- 2) STAT BIOS.SYS \$R/W (the file is normally write protected)
- 3) DDT BIOS.SYS (call BIOS.SYS into memory)
- 4) SOCF1 (cr) (select patch mode and starting address)
- 5) 20 (cr) (this is to change value from 10H to 20H)
- 6) CA (cr) (to change JNZ to JZ)
- 7) . (cr) (this exits the 'S' mode)
- 8) CNTRL C (to exit DDT)
- 9) SAVE 18 BIOS.SYS (to write BIOS.SYS back to disk)
- 10) STAT BIOS.SYS \$R/O (write protect file)
- 11) STAT BIOS.SYS \$SYS (prevent file from showing up on DIR command)

Tor says the easier way to do this is get Terry Bolstad's DDUMP program (see last month H8SCOOP). It does the same for CP/M as DUMP does for HDOS and is highly recommended for CP/M users.

MX PRINTER HDOS-CP/M

While on the subject, Charles Bracher/ 6221 Fenley Dr/ Huntington Beach, CA 92647 sends along these ideas on interfacing the Epson printers under HDOS and CP/M. He has an Epson MX-100, and is using Lindley's HDOS device driver. He is running the Magnolia CP/M and had no drivers for the Epson. Here's the way he gets the MX-100 operating under CP/M.

"At the present time I get the MX-100 operating OK by first configuring the printing by using HDOS and Microsoft and then re-boot into Magnolia's CP/M. As long as one does not turn off the printer, it will retain the commands received from the Microsoft routine. Incidentally, I am using the regular Heath interface cable for the MX-100. When I got to Magnolia CP/M, I just jumper pins 4 and 11 on the printer RS-232 connector and the printer works fine on CP/M. This might help some Magnolia users who probably have to use two cables."

BOOT SY1: WITH MTR-88 ROM

From David Green, a way to boot with the old monitor ROM from SY1:. Using the substitute command (H89) type in the following, then start the program at 042200A with the GO command. The new monitor ROM's allow booting from other drives than SY0:, so this would not be needed.

042.200	STACK	EQU	42200	
000.130	BOOTAL	EQU	130A	
037.132	BOOTA	EQU	37132A	
040.110	D.CON	EQU	40110A	
030.252	\$MOVE	EQU	30252A	
031.212	\$ZERO	EQU	31212A	
040.240	D.RAM	EQU	40240A	
000.037	D.RAML	EQU	037Q	
041.061	AIO.UNI	EQU	41061A	
042.200		ORG	42200A	
042.200	363	DI		
042.201	061	200	042	LXI SP,STACK
042.204	001	130	000	LXI B,BOOTAL
042.207	021	132	037	LXI D,BOOTA
042.212	041	110	040	LXI H,D.CON
042.215	315	252	030	CALL \$MOVE
042.220	041	240	040	LXI H,D.RAM
042.223	006	037		MVI B,D.RAML
042.225	315	212	031	CALL \$ZERO
042.230	076	001		MVI A,1
042.232	062	061	041	STA AIO.UNI
042.235	257			XRA A
042.236	303	047	037	JMP 37047A

HUG DUMP WITH 400K DRIVES

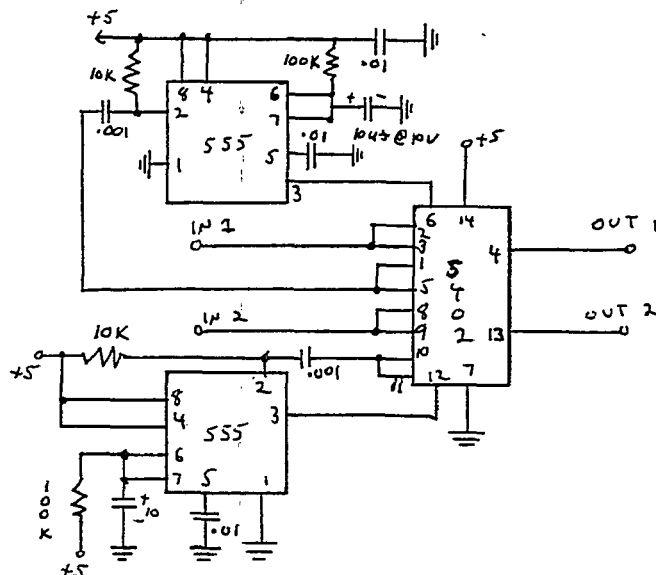
Dave Green also says HUG's DUMP utility works in filemode with the Tandon TM-100-4 400K drive by patching with PATCH. Change address 44066 from 050 to 240. This changes CPI 40 track to CPI 160 track. There are other changes for cosmetic purposes, if you want these, or have questions on the above, contact David Green/ 2390D Columbian Ct/ Birmingham, AL 35216. Dave also says the Software Toolworks program SZAP works with the new drives in filemode but does not work in the raw track/sector mode.

DISK DRIVE HEAD UNLOAD DELAY

This information came in from Larry Reeve of Polybytes/ 325 19th St. S.E./ Cedar Rapids, IA 52403. He writes concerning the new SY.DVD driver from UltiMeth "I am impressed with the new driver with the exception that I am disappointed with the method of delaying the head unloading. The method he used results in the wait for load to be of the same duration as the wait to unload (could the be the reason the DUP programs must have SELECT set to 0?--ed). If you set the SELECT delay to be around .75 seconds, a very comfortable time, there is always a 3/4 second delay between file accesses. When I installed the new driver I cut out my delay circuit, but unless I can figure out the proper software modifications, I'll be re-installing it."

His presented delay circuit can be installed in the drive by opening the HS jumper, connecting the IN point on the schematic to the computer side of the HS jumper, and connecting the OUT point to the drive side of the HS jumper. The circuit will provide a 3/4 second delay on

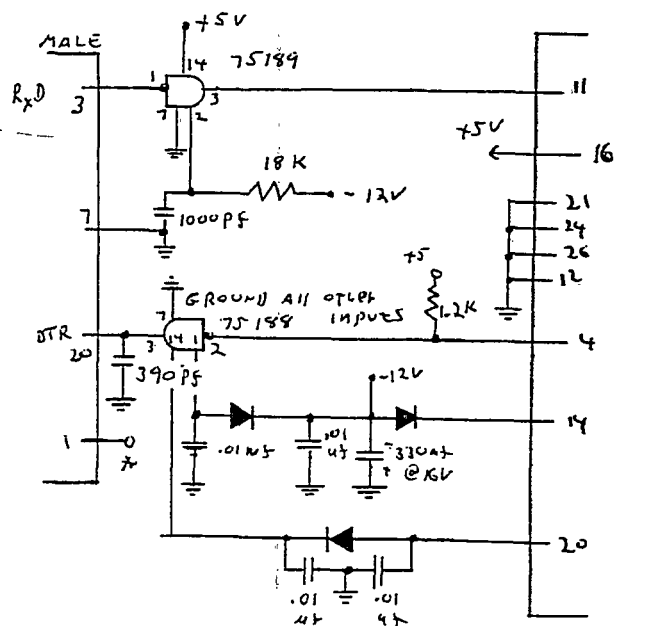
head unload for two drives (good for the old 100K drives to eliminate much of the head load/unload noise). This circuit has been used for over a year without any problems.



911 "C" in 45

MX-80 INTERFACE CIRCUIT

Larry also sends along a circuit he uses for the MX-80. The circuit can be installed inside the MX-80 in the position designed for the serial interface. It will communicate at 9600 baud with 8 data bits and no parity. There is an HDOS 2.0 device driver that goes along with it. If anyone wants it, contact Larry.



* connect to logged with presently attached to main logic board

INTERNAL MX-80 CONNECTOR

CLASSIFIEDS

Classified policy is simple and as follows. Subscribers allowed free advertising, non subscribers pay \$5 per insertion.

FOR SALE--Heavy Duty Rack mounted power supplies. 5V @ 20A adjustable with overvoltage protection, +/- 12 or +/- 15V adjustable at 4A regulated, 24V unregulated, -6V unregulated. Powerfail line. Ideal for powering microcomputers or disk drive units or whatever. Steal at \$95, limited quantity. Contact H8SCOOP.

FOR SALE--One HP 2621A video display terminal in original box with all manuals included. This is a professional unit with 2 pages of memory, scroll up/down, previous/next page, addressing, editing, N-key rollover, auto repeat any key, detached keyboard, capable of displaying control characters as a selectable mode. This is an extremely nice terminal in mint condition. I pay shipping. \$1095. Contact Brian Branson/ 2255 Cahuilla Rd Colton, CA 92324. Phone (714) 824-0144.

FOR SALE--2 Godbout 12K static memory boards for the H8. \$150 each, or \$275 for the pair. Contact Howard Rifkin/ 8 Harold St/ Old Bridge, NJ 08857. Phone (201) 679-6494.

WANTED--Z89 computers (used-below \$1500) at a reasonable cost. Contact Allen Bernson/ POB 68/ Boston, MA 02167.

FOR SALE--IBM Model 735 I/O printer. Machine type 1011 Mag tape set Typewriter Mod 2 775 Printer 10 pitch, Fabric ribbon. 15" carriage. 24 Volt solenoids, standard electric keyboard. 12 years old, but used only 2 years. Very clean. \$675. Stanley Dhein/ Dhein's Office Machines/ 1055 Elm St/ Cleveland, WI 53015. Phone (414) 693-8760.

FOR SALE--DG-32 32K Dynamic RAM for the H8, \$275. Heath 8K Static RAM for the H8 for \$60. Will sell both for package price of \$300 for the 40K. Dale Grant. Phone (513) 426-0896.

WANTED--Large quantities of 8K memory boards for the H8. Must be in excellent condition. **FOR SALE**--LA36 Decwriter II for \$600. Contact U.S. Energy/ 78-40 164th St/ Flushing, NY 11366/ (212) 380-1004.

FOR SALE--H8 computer, one year old with 56K RAM, H17 dual floppy, H19 video terminal, H8-4 interface, assorted software, blank disks and manuals. Asking price \$2,000 or best offer. Contact Terrance Brodie/ POB 396/ Algonquin, IL 60102. Phone (312) 658-3661.

FOR SALE--2 Heath 8K RAM boards for \$50 each. David Carrick/ 1960 Hunt Club Dr/ Goose Pointe Woods, MI 48236. Phone (313) 884-0781.

PEEKING & POKING

ETHICS

I will attempt to keep this as brief as possible since lot of you do not care for the editorial section, but I feel in this instance, since it touches each and every one of us, it would be wise to think about and ponder these words.

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Back issues #'s 1, 2, 3, 4, 6, 7, 8, 9, 10, 11 no longer available. Remainder through March 1981 (issue 12) still available at \$1 each while supply lasts. Volume 2 back issues available at \$1.25 each to current issue.

PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday. NO COLLECT CALLS.

I know for a fact that most of you that have software did not buy the majority of it. It has been passed along, or traded, or sold for a good price through a friend. That's all well and good to a point, and I'll explain myself on that. I get hundreds of discs every few months, and non of these I pay for. However most of these are for evaluation purposes, and if using one that the author did not directly send me I usually end up buying it anyway--it's only fair. One good example is the software I am now using to prepare this newsletter, SPELLBINDER. It was given to me, and I like it so much I am going to buy a copy, just to keep things straight. WORDSTAR also passed my desk, and there's no way I would shell out money for that. This is true of lot of software. I feel why purchase what you don't know the value (to you) of. But if you get something and like it, then by all means buy it!

We think computers are so great and laugh when we get a good deal on software which was given to us or traded. However, what does the author get for it? Great big goose egg zero, you guessed it. Now the author most likely has a family to support and depends on his sales for income. He may be doing this for a full time living trying to make a go of it, or he may be getting to the point of doing it full time, yet still has to hold down another job because his sale income is not yet great enough. Enough of these individuals can be seen in the industry today that are exactly in one or the other situation. I'm a good example of that (and I don't enjoy having my newsletters given away, nor my OFFERS), so is Walt Bilofsky, Ted Benglen, Don Long, and many more.

I heard a good but sad story from Bill Parrot, software developer for D-G Electronics. He was at a Heath store and right before his eyes, some of HIS software was being sold or traded--xerox copies of the manuals and disk copies, software he would never receive any money for! It made Bill's day.

I have heard comments about my disc offers inferring I don't mind if one person buys them and gives them away or trades them. I DO mind. The persons who send me the discs and myself put lot of time and development into these products. They receive some compensation, and I use the profit to keep my equipment going and keep the price of H8SCOOP reasonable. When the price finally jumps up with all the others (BUSS is now \$25/year) then blame yourselves!

The point? The point is that if we continue to support these guys and show them that we care and they can make a living there will be a lot more quality products and a lot more people cranking it out. This means for you better products for lower cost. Let's face the facts, the majority of good hardware and software for your systems is NOT coming from Heath, but from outside sources!

If you want to cheat the authors and get your software free fine. The'll eventually be less and less quality software out there, and less and less support, and then you'll all be sitting and complaining. It ALWAYS turns out the ones that buy the least (cheat the most) complain the loudest when things go wrong! If this trend continues, we could find lot of the good guys out there dropping out. It's only fair if you like a product and use it, then pay for it. What the future holds is totally up to you. Think about it and ponder it carefully, for you hold the future in your hands.